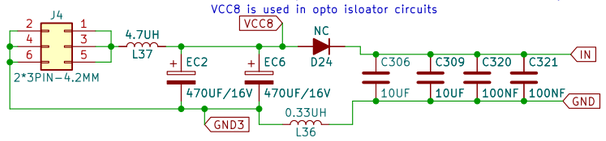
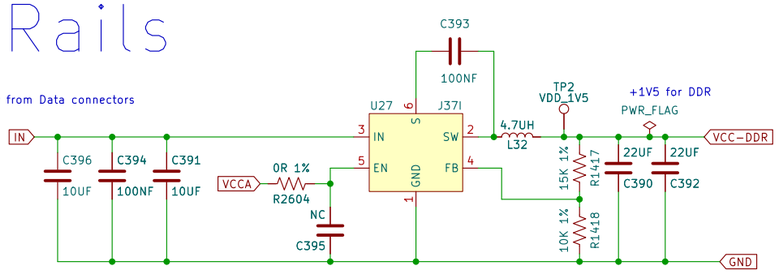
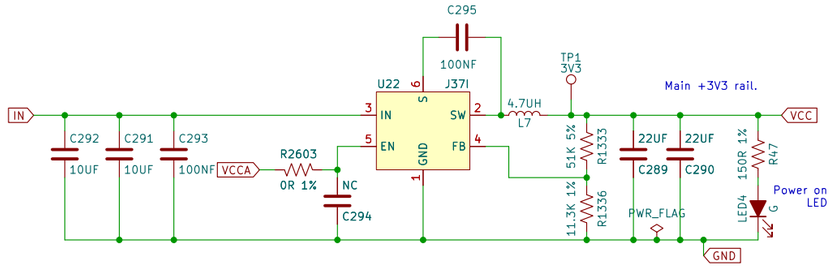
Power Rails

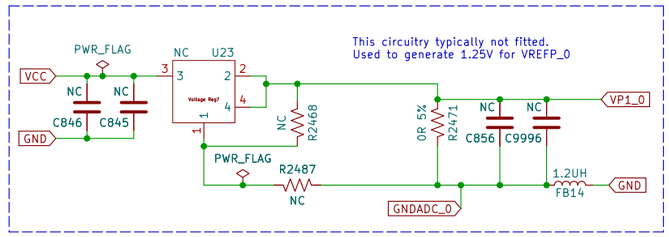
Main Power connector con only be used if D24 is fitted.
VCC8 is used in opto isloator circuits

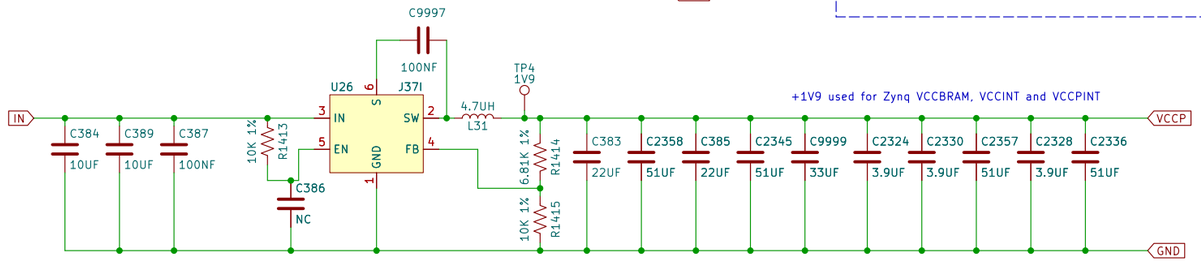


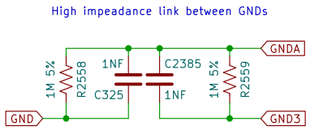
Injection point for power from Data connectors

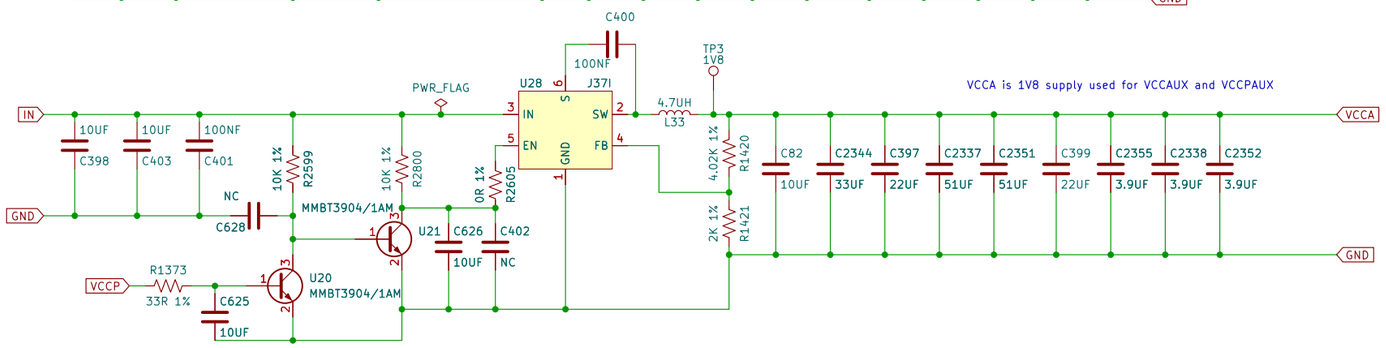












R9999 was R120A
R9998 was R2445B
R9997 was R2473A
R9996 was R2487A
R9995 was R2641A

C9999 was C2349A
C9998 was C379A
C9997 was C386A
C9996 was C856A

Kicad does not allow refs like RxxxA some some devices have been renumbered
Extra PWR_FLAGS have been added to clear DRC errors
Some schematic symbols have been eeditted to clear DRC errors

Sheet: /
File: ebaz4205.kicad_sch

Title: EBAZ4205 Schematic rebuild.

Size: A3	Date: 2021-02-15	**Rev: V004**
KiCad E.D.A. 10.0.6		Id: 6/6

Ethernet PHY

U24A
IP101GA

VDDIO
AVDD33
DGND
DGND
AGND1V
AGND33
DGND
ISET
X1
X2
Power
GND
Crystal

C369 C370 C374 C367 100NF C9998
R1393 6.2K 1%
R1485 NC
XOUT.0_0
Early revs of build have Xtal and caps fitted, but not R1485
Later builds have R1485 (OR) but not caps & Xtal
C377 22PF C380 22PF
Y3 25.000MHZ

U24D
IP101GA
R1381 5.1K 1%
REGOUT.0
NC

U24B
IP101GA
RXCLK_0 RXD0_0 RXD1_0 RXD2_0 RXD3_0 RXDV_0
R1410 R1391 R1394 R1397 R1405 R1385 R2560 R1396 R1408
RXCLK/50M_CLKO
RXD0 RXD1 RXD2 RXD3
RXDV/CRS_DV/FX_HEN
RXER
CRS/LEDMOD
COL/RMII
RX MII
TXCLK_0 TXD0_0 TXD1_0 TXD2_0 TXD3_0 TXEN_0
TXCLK/50M_CLKI
TXD0 TXD1 TXD2 TXD3 TXEN
TXER/FXSD
TX MII
MDC_0
MDC MDIO
REST.ON_0
TEST_ON
Test
RESET_N
Reset
INTR
Interrupt
R2470 R2469 10K 1% R1995
RESET.1
C376 100NF

U24C
IP101GA
MDI_RN MDI_RP MDI_TN MDI_TP
RX In Pair
TX Out Pair
REGOUT
LED0/PHY_AD0
LED1/PHY_AD1
LED2/PHY_AD2
LED3/PHY_AD3
LED /Addr
LED.0_0 LED3_0
R1388 R1411 R1406 R1409 C366 C372
5.1K 1% R1390 R1392 R1386 R1383

BT16B03
RX− RXCT RX+ NC NC TX− TXCT TX+
RD− RCT RD+ NC NC TD− TCT TD+
T6

U25 RJ45
LED3_0 LED.0_0
R1407 R1404 R1384 R1400 75R 1%
330R 1% R1389 R1412 100NF C659 C660

REGOUT REGOUT.0
NC L30
C382 NC C379 C373 C375 C371 C368 100NF

Typically not fitted

Not sure what this device is. Maybe alternative reset supervisor?
U30 ACOI
PWR_FLAG
R1482 3K 5% IN
R1483 1.2K 5% GND
R1481 33R 1% U31.W20_0

Reset Supervisor

U65 SGM706-SYS8
C1887 100NF
R1949 NC 100K 1% R1948
10K 1% R2336
10K 1% R1950
R1401 33R 1% RESET.1
R2392 33R 1% RESET.0

JTAG Connector

The JTAG connector is not fitted on the later builds.
J8
0R 1% R2602 4.7K 1% R38 4.7K 1% R1507 R36 4.7K 1%
22R 1% R1504 U31-TMS_0
22R 1% R37 U31-TCK_0
22R 1% R1505 U31-TDO_0
22R 1% R1506 U31-TDI_0

RTC, This circuitry is typically not fitted. Exact devices have not been identified.

Y2 4pin SMD NC
X3 DS1307
R1279 NC
C68 NC C538 NC C71 NC C69 NC
R2440 NC R117 NC
R120 NC R2438 NC
R122 NC R2439 NC
U31-A15_0 U31-D13_0

Kicad does not allow refs like RxxxA some some devices have been renumbered
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Some schematic symbols have been eeditted to clear DRC errors

Sheet: /PHY & Sundries/
File: Phy-Sun.kicad_sch

Title: EBAZ4205 Schematic rebuild.

Size: A3	Date: 2021-02-15	Rev: V004
KiCad E.D.A. 10.0.6		Id: 2/6

Flash

U12
W29N01HVSINA

R1358 4.7K 1% RY/#BY_0
R1359 4.7K 1% #CE_0
R2583 NC #RE_0
R2590 20K 1%
R2582 NC CLE_0
R2580 NC ALE_0
R2579 NC #WE_0
R1360 4.7K 1%
R2586 20K 1%
R2587 20K 1%
R2598 20K 1%
R244 NC
C670 3.9UF C673 3.9UF C669 51UF
R245 NC
R247 NC
R2578 20K 1%
R2581 NC
R2577 NC
R248 NC
R2584 20K 1%
R2588 20K 1%
R2585 NC
R246 0R 1%

MIO4
MIO5

	BOOT MODE	
	IO0	IO2
MIO	5	4
JTAG	0	0
NAND	0	1
SD	1	1

DDR Termination, This circuitry is typically not fitted. Exact devices have not been identified.

U72
TPS51200

C2397 NC C2396 NC C2399 NC R2615 NC R2616 NC
C2395 NC C2401 NC R2618 NC
C2400 C2398 3.9UF C2394 100NF
R2619 NC R2617 NC

DDR

U66A
EM6GD16EWKG-12H

U66B
EM6GD16EWKG-12H

C2300 100PF C2302 100PF R2537 1.2K 1% R2538 1.2K 1% C2383 100NF C2384 100NF PWR_FLAG
R2542 80.5R 1%
R2645 4.7K 1%
R2455 240R 1%

U31E
XC7Z010-1CLG400I

Data Bus to Flash

U31G
XC7Z010-1CLG400I

PWR_FLAG
1.2UH L113 NC R2391
100PF C2285 100PF C2287 R2389 NC
X8 33.333MHZ
R2340 NC CLK_0

Used for DDR termination. Normally not fitted
C2402 C2409 C2404 C2403 C2406 C2407 C2405 C2408

R2457 80.5R 1%
C2382 10NF R2458 1.2K 1% C2303 100NF R2459 1.2K 1% C2304 100NF
R2159 1.2K 1% C1985 100NF R2161 1.2K 1% C1987 100NF R2456 80.5R 1%

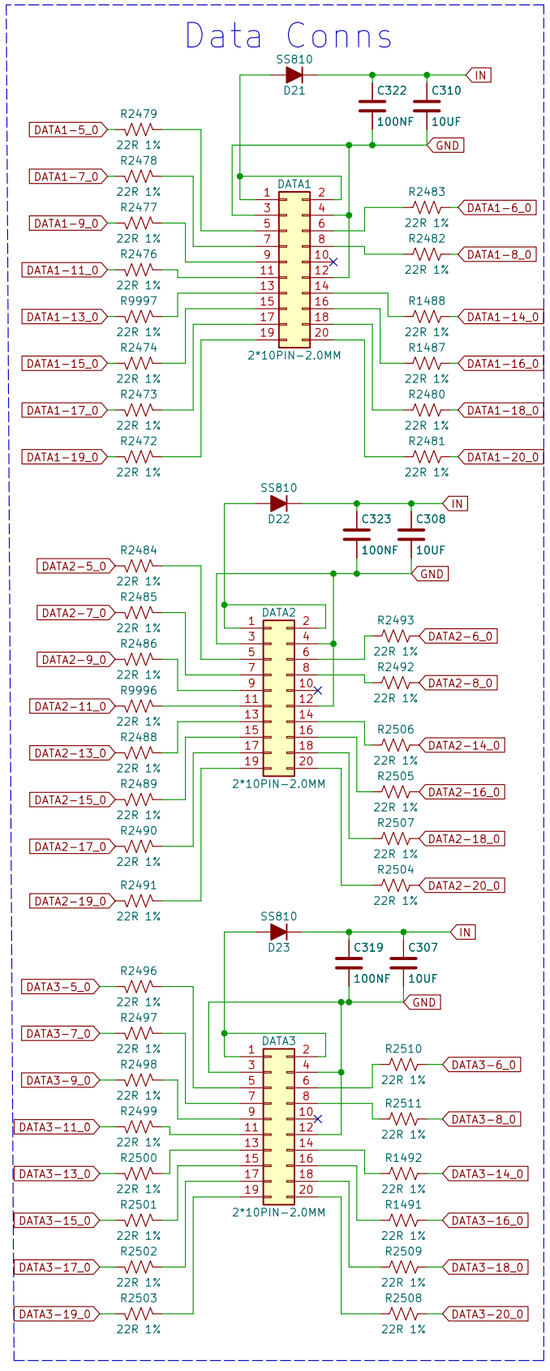
R2621–R2638 NC DDR-A0_0–DDR-A14_0 R2517–R2535 10R 1% PS-DDR-A0_0–PS-DDR-A14_0
R2642 NC DDR-ODT_0 R2513 10R 1% PS-DDR-ODT_0
R2639 NC DDR-RAS_0 R2532 10R 1% PS-DDR-RAS_0
R2640 NC DDR-CAS_0 R2533 10R 1% PS-DDR-CAS_0
R2644 NC DDR-CKE_0 R2536 10R 1% PS-DDR-CKE_0
R2643 NC DDR-CS_0 R2535 10R 1% PS-DDR-CS_0
R2641 NC DDR-WE_0 R2534 10R 1% PS-DDR-WE_0
R2636 NC DDR-BA0_0 R2514 10R 1% PS-DDR-BA0_0
R2637 NC DDR-BA1_0 R2515 10R 1% PS-DDR-BA1_0
R2638 NC DDR-BA2_0 R2516 10R 1% PS-DDR-BA2_0

Kicad does not allow refs like RxxxA some some devices have been renumbered
Extra PWR_FLAGS have been added to clear DRC errors
Some schematic symbols have been eeditted to clear DRC errors

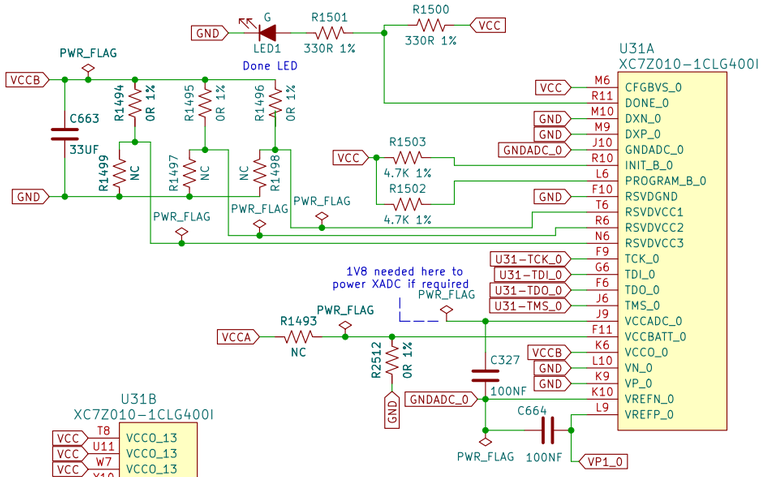
Sheet: /Mem & Zynq Sheet/
File: Mem_Zynq.kicad_sch

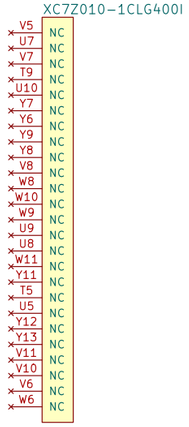
Title: EBAZ4205 Schematic rebuild.

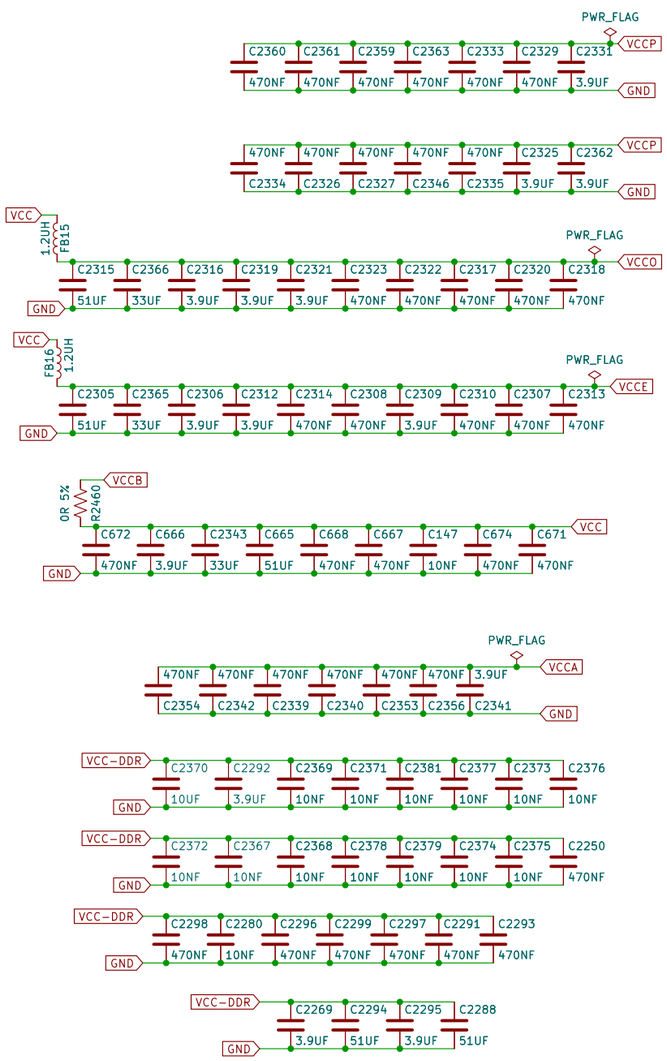
Size: A3 | Date: 2021-02-15 | **Rev: V004**
KiCad E.D.A. 10.0.6 | Id: 3/6

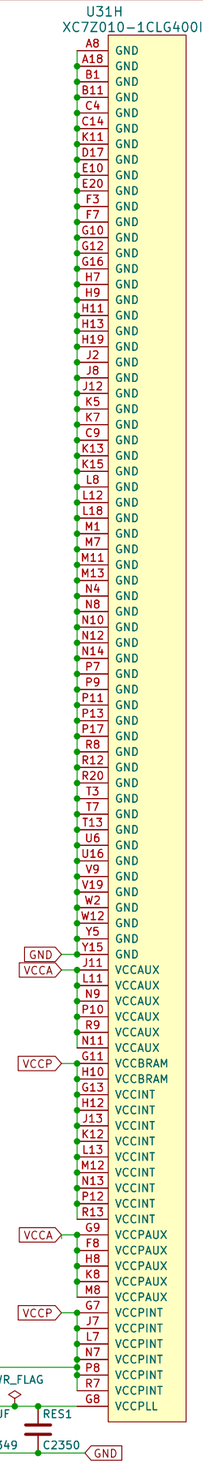

Data Conns
SS810
D21
C322
100NF
C310
10UF
IN
GND
DATA1
2*10PIN-2.0MM
R2479
R2478
R2477
R2476
R9997
R2474
R2473
R2472
R2483
R2482
R2488
R2487
R2480
R2481
DATA1-5_0
DATA1-7_0
DATA1-9_0
DATA1-11_0
DATA1-13_0
DATA1-15_0
DATA1-17_0
DATA1-19_0
DATA1-6_0
DATA1-8_0
DATA1-14_0
DATA1-16_0
DATA1-18_0
DATA1-20_0
22R 1%
SS810
D22
C323
100NF
C308
10UF
IN
GND
DATA2
2*10PIN-2.0MM
R2484
R2485
R2486
R9996
R2488
R2489
R2490
R2491
R2493
R2492
R2506
R2505
R2507
R2504
DATA2-5_0
DATA2-7_0
DATA2-9_0
DATA2-11_0
DATA2-13_0
DATA2-15_0
DATA2-17_0
DATA2-19_0
DATA2-6_0
DATA2-8_0
DATA2-14_0
DATA2-16_0
DATA2-18_0
DATA2-20_0
22R 1%
SS810
D23
C319
100NF
C307
10UF
IN
GND
DATA3
2*10PIN-2.0MM
R2496
R2497
R2498
R2499
R2500
R2501
R2502
R2503
R2510
R2511
R1492
R1491
R2509
R2508
DATA3-5_0
DATA3-7_0
DATA3-9_0
DATA3-11_0
DATA3-13_0
DATA3-15_0
DATA3-17_0
DATA3-19_0
DATA3-6_0
DATA3-8_0
DATA3-14_0
DATA3-16_0
DATA3-18_0
DATA3-20_0
22R 1%
PWR_FLAG
VCCE
NC
L29
C363
C362
C364
GND
X5
EN
VDD
OUT
GND
R277
R1372
U31C
XC7Z010-1CLG400I
R2540
R2541
20K 1%
OPTO J3P4 OUT
OPTO J3P3 IN
GRN LED6
OPTO J5P4 OUT
RED LED6
OPTO J5P3 IN
R2461
22R 1%
TXCLK_0
PHY CLK
XOUT.0_0
DATA3-14_0
DATA3-18_0
U3.U12_0
U3.V13_0
U3.W13_0
U3.V12_0
U3.W14_0
U3.V15_0
MDC_0
RXD3_0
RXD0_0
REST.ON_0
RXCLK_0
DATA3-20_0
DATA3-15_0
DATA3-11_0
DATA3-6_0
DATA3-17_0
DATA3-16_0
U31.W20_0
DATA3-19_0
IO_0_34
IO_25_34
IO_L1N_T0_34
IO_L1P_T0_34
IO_L2N_T0_34
IO_L2P_T0_34
IO_L3N_T0_DQS_34
IO_L3P_T0_DQS_PUDC_B_34
IO_L4N_T0_34
IO_L4P_T0_34
IO_L5N_T0_34
IO_L5P_T0_34
IO_L6N_T0_VREF_34
IO_L6P_T0_34
IO_L7N_T1_34
IO_L7P_T1_34
IO_L8N_T1_34
IO_L8P_T1_34
IO_L9N_T1_DQS_34
IO_L9P_T1_DQS_34
IO_L10N_T1_34
IO_L10P_T1_34
IO_L11N_T1_SRCC_34
IO_L11P_T1_SRCC_34
IO_L12N_T1_MRCC_34
IO_L12P_T1_MRCC_34
IO_L13N_T2_MRCC_34
IO_L13P_T2_MRCC_34
IO_L14N_T2_SRCC_34
IO_L14P_T2_SRCC_34
IO_L15N_T2_DQS_34
IO_L15P_T2_DQS_34
IO_L16N_T2_34
IO_L16P_T2_34
IO_L17N_T2_34
IO_L17P_T2_34
IO_L18N_T2_34
IO_L18P_T2_34
IO_L19N_T3_VREF_34
IO_L19P_T3_34
IO_L20N_T3_34
IO_L20P_T3_34
IO_L21N_T3_DQS_34
IO_L21P_T3_DQS_34
IO_L22N_T3_34
IO_L22P_T3_34
IO_L23N_T3_34
IO_L23P_T3_34
IO_L24N_T3_34
IO_L24P_T3_34
VCCO_34
Data to Eth Phy
R2466
R2464
R2465
R2462
R2463
TXD3_0
TXD1_0
TXD2_0
TXEN_0
TXD0_0
RXDV_0
RXD1_0
RXD2_0
DATA3-13_0
DATA3-7_0
DATA3-9_0
U31D
XC7Z010-1CLG400I
DATA1-8_0
DATA1-9_0
DATA1-5_0
DATA1-7_0
DATA1-14_0
DATA1-13_0
DATA1-16_0
DATA1-18_0
DATA2-19_0
DATA3-5_0
DATA2-17_0
DATA3-8_0
DATA2-18_0
DATA2-16_0
DATA2-9_0
DATA2-13_0
DATA2-20_0
DATA2-15_0
DATA2-11_0
DATA1-20_0
DATA1-11_0
DATA1-6_0
DATA1-15_0
DATA2-6_0
DATA1-17_0
DATA1-19_0
DATA2-8_0
DATA2-14_0
DATA2-5_0
DATA2-7_0
?? U31-N16_0
?? U31-L14_0
IO_0_35
IO_25_35
IO_L1N_T0_AD0N_35
IO_L1P_T0_AD0P_35
IO_L2N_T0_AD8N_35
IO_L2P_T0_AD8P_35
IO_L3N_T0_DQS_AD1N_35
IO_L3P_T0_DQS_AD1P_35
IO_L4N_T0_35
IO_L4P_T0_35
IO_L5N_T0_AD9N_35
IO_L5P_T0_AD9P_35
IO_L6N_T0_VREF_35
IO_L6P_T0_35
IO_L7N_T1_AD2N_35
IO_L7P_T1_AD2P_35
IO_L8N_T1_AD10N_35
IO_L8P_T1_AD10P_35
IO_L9N_T1_DQS_AD3N_35
IO_L9P_T1_DQS_AD3P_35
IO_L10N_T1_AD11N_35
IO_L10P_T1_AD11P_35
IO_L11N_T1_SRCC_35
IO_L11P_T1_SRCC_35
IO_L12N_T1_MRCC_35
IO_L12P_T1_MRCC_35
IO_L13N_T2_MRCC_35
IO_L13P_T2_MRCC_35
IO_L14N_T2_AD4N_SRCC_35
IO_L14P_T2_AD4P_SRCC_35
IO_L15N_T2_DQS_AD12N_35
IO_L15P_T2_DQS_AD12P_35
IO_L16N_T2_35
IO_L16P_T2_35
IO_L17N_T2_AD5N_35
IO_L17P_T2_AD5P_35
IO_L18N_T2_AD13N_35
IO_L18P_T2_AD13P_35
IO_L19N_T3_VREF_35
IO_L19P_T3_35
IO_L20N_T3_AD6N_35
IO_L20P_T3_AD6P_35
IO_L21N_T3_DQS_AD14N_35
IO_L21P_T3_DQS_AD14P_35
IO_L22N_T3_AD7N_35
IO_L22P_T3_AD7P_35
IO_L23N_T3_35
IO_L23P_T3_35
IO_L24N_T3_AD15N_35
IO_L24P_T3_AD15P_35
VCCO_35
Kicad does not allow refs like RxxxA some some devices have been renumbered
Extra PWR_FLAGS have been added to clear DRC errors
Some schematic symbols have been eeditted to clear DRC errors
Sheet: /Zynq_IO Sheet/
File: Zynq_IO.kicad_sch
Title: EBAZ4205 Schematic rebuild.
Size: A3
Date: 2021-02-15
Rev: V004
KiCad E.D.A. 10.0.6
Id: 4/6

Zynq Power









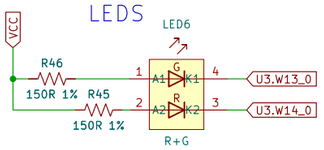
Kicad does not allow refs like RxxxA some some devices have been renumbered
Extra PWR_FLAGS have been added to clear DRC errors
Some schematic symbols have been eeditted to clear DRC errors

Sheet: /Zynq Power Sheet/
File: Zynq_Pwr.kicad_sch

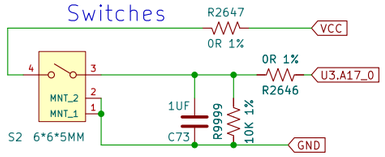
Title: EBAZ4205 Schematic rebuild.

Size: A3	Date: 2021-02-15	**Rev: V004**
KiCad E.D.A. 10.0.6		Id: 5/6

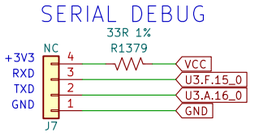
LEDS

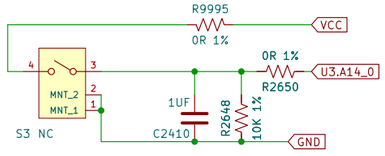


Switches



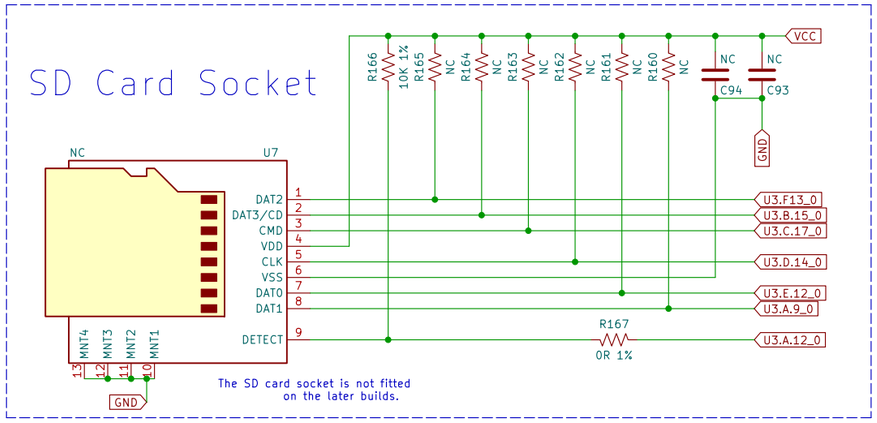
SERIAL DEBUG



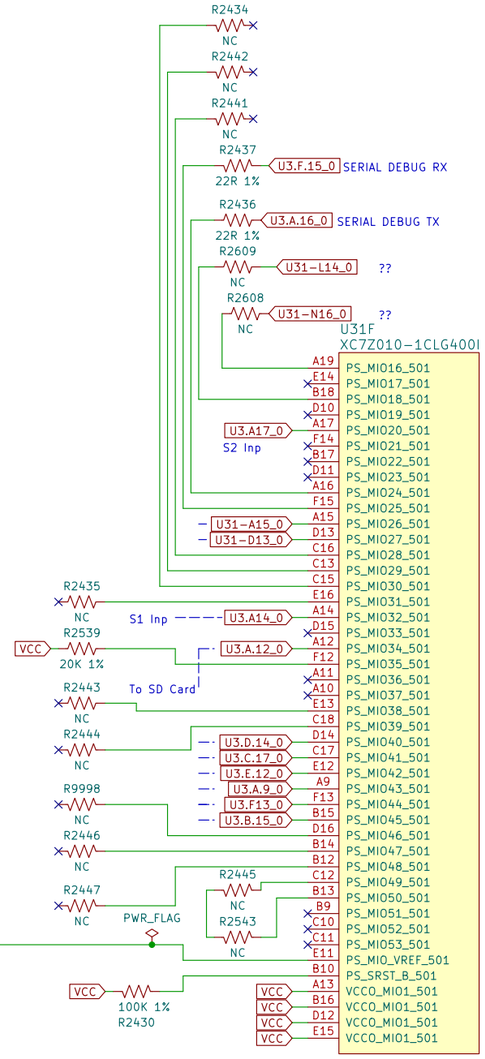


S3 typically not fitted

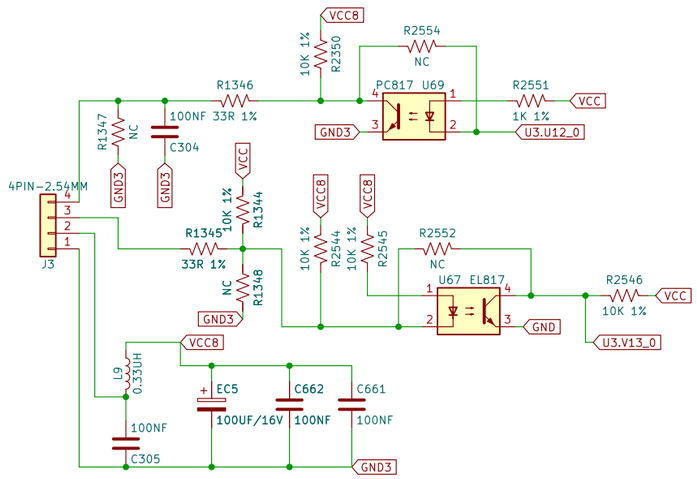
SD Card Socket



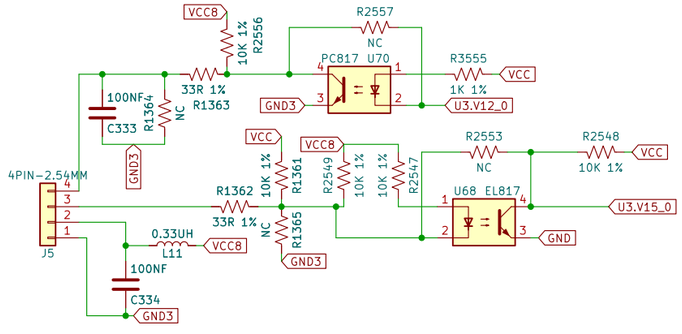
The SD card socket is not fitted on the later builds.



Opto-Isolators



Later build boards do not have the opto devices fitted.
In this case the bypass resistors are fitted to allow a single input and output on both J3 and J5



Kicad does not allow refs like RxxxA some some devices have been renumbered
Extra PWR_FLAGS have been added to clear DRC errors
Some schematic symbols have been eeditted to clear DRC errors

Sheet: /IO Sheet/
File: IO.kicad_sch

Title: EBAZ4205 Schematic rebuild.

Size: A3	Date: 2021-02-15	Rev: V004
KiCad E.D.A. 10.0.6		Id: 6/6